



React Redux

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1. What is redux

- React Redux is a powerful state management library for React applications.
- It provides a centralized store to manage your application's state,
- Without Redux, it is required to make data dependent on the components and pass it through different components to where it's needed.

2. Advantages of redux

- Centralized state management: state of your entire application in a single store
- Predictable state updates: data in the store gets updated in response to user actions.
- Easier debugging: Different Redux DevTools are available
- Better performance: Redux can minimize unnecessary re-renders of components.

- All components in your application can access the central store and update or retrieve data from it as needed.
- The key components that enable this centralized approach to state management are:
 - Store
 - Actions
 - Dispatch
 - Reducers
- Store:
 - The Redux store is like a giant container that can hold any kind of data required in the application.
 - Any component in the application can access it to retrieve and update data.

- Actions
 - An action is an object having type and payload property
 - A type property is a string constant describing the action
 - The payload property represents the data required by the action
- Dispatch
 - It is a function provided by the store to send an action to update the state
- Reducers
 - A reducer is a function that takes in the current state of an application and an action as arguments, and returns a new state

4. Steps for implementation

1. Install latest redux library
2. Create store by using `configureStore()` function
3. Wrap `<App />` component in `<Provider />` element
4. Create slice by using `createSlice()` function needs object as argument which has keys - `name`, `initialState(Object)`, `reducers(object)`.
5. Once the slice is ready, make the entry in store
6. Access state in the component by using `useSelector()` hook.
7. Use `useDispatch` for modifying the state